Shesadree Priyadarshani

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EDUCATION

University of Florida — **Aug 2025 – May 2027**
M.S. Applied Data Science, GPA: 4.0/4.0 — *Gainesville, FL*
Coursework: Advanced Machine Learning, Statistical Inference, Applied Data Science, AI Ethics, Advanced Mathematics

National Institute of Technology Rourkela — **Aug 2019 – May 2023**
B.Tech. Electrical Engineering — *Rourkela, India*

RESEARCH & EXPERIENCE

UF Intelligent Clinical Care Center (IC3), University of Florida — **Nov 2025 – Present**
AI Research Assistant – I-HEAL Research Group — *Gainesville, FL*

- Profiled transformer inference on **HiPerGator accelerators** using **PyTorch and CUDA**, characterizing **KV-cache memory growth**, prompt-processing versus token-generation latency, and GPU utilization under varied sequence and micro-batch configurations; evaluated prefix sharing for recurring clinical workloads.
- Developed reproducible **zero-shot and few-shot evaluations** for clinical prediction, covering accuracy, AUROC, calibration, bootstrap confidence intervals, and error cohorts; accuracy by **14%** and AUROC by **9%**.
- Integrated **RAG and QLoRA fine-tuning** with structured retrieval, embedding search, prompt iteration, and grounding analysis; improved grounding by **30%**, reduced hallucinations by **35%**, and raised reasoning consistency by **18%**.
- Prototyping a **mechanistic-neural ODE** model for ICU outcomes that combines physiological priors, neural residuals, and graph-structured clinical data, with sensitivity and feature-attribution methods for debugging and interpretability.

Engineering Education Department, University of Florida — **Jun 2026 – Present**
Graduate Student Assistant — *Gainesville, FL*

- Created an end-to-end **agentic AI system** with Claude and **MCP connectors**, combining structured ingestion, multi-step synthesis, validation checks, and an interactive Voilà dashboard for alumni and faculty intelligence.

Bristol Myers Squibb — **Apr 2024 – Jul 2025**
Decision Analyst I – Data Science and AI Solutions — *Hyderabad, India*

- Engineered reusable **Python feature generation and batch scoring** across **5+ global data sources** for forecasting, retention, and sentiment models; automated quality checks and KPI reporting, improving forecast accuracy by **15%** and supporting **$2M+** incremental ROI.
- Led offline model assessment and **A/B/test-control experiments**; applied cohort analysis and statistical testing to isolate performance drivers and convert findings into decisions for cross-functional teams.

ZS Associates — **Jul 2023 – Apr 2024**
Decision Analytics Associate – Multi-Channel Marketing — *Pune, India*

- Developed **SQL, R, and Python feature engineering** and predictive analyses for response and segment performance; standardized model-ready datasets and recurring scoring jobs, reducing refresh time by **30%** and manual effort by **20%**.

SELECTED PROJECTS

Distributed Speculative Decoding and Prefix-Cache Serving | PyTorch, CUDA, Ray, FastAPI — **Jul 2026**

- Implementing a distributed LLM serving system with **Ray-managed draft workers**, target-model verification, prefix-aware state reuse, CUDA-optimized batching, and reproducible multi-user load generation.
- Evaluating **time-to-first-token, token-generation latency, p50/p95 latency, throughput, speculative acceptance rate, GPU utilization**, and output parity against standard autoregressive decoding.

Stateful Inference and Cache Policy Lab | PyTorch, FastAPI, Redis, Docker — **Jun 2026**

- Developing a stateful inference service that reuses encoder and prefix computations through content-addressed embedding caches, request deduplication, and reusable model states exposed through a containerized **FastAPI** interface.
- Comparing **LRU, TTL, and hybrid eviction policies** under bursty traffic and distribution drift while tracking cache hit rate, p50/p95 latency, throughput, eviction behavior, and stale-reuse frequency.
- Designing a correctness-risk score from **cosine/L2 embedding divergence and output-parity checks** to quantify stale-cache impact and trigger safe recomputation when reuse becomes unreliable.

ECG Arrhythmia Classification | Python, PyTorch, 1D-CNN, LSTM — **May 2026**

- Trained a leakage-safe model on **MIT-BIH ECG signals**, combining learned waveform representations, RR-interval rhythm context, and class-weighted loss across five AAMI heartbeat classes.
- Reached **0.91 recall** on ventricular arrhythmias and deployed an interactive **Streamlit** demo for waveform visualization, class probabilities, and error analysis.

TECHNICAL SKILLS

Languages & Systems: Python, SQL, R, CUDA, Linux, Git, REST APIs, FastAPI, HiPerGator HPC
ML Systems: PyTorch, Hugging Face Transformers, LLM inference, KV caching, speculative decoding, GPU profiling, batching, distributed serving, Ray
Generative AI & Evaluation: RAG, QLoRA, LangChain, LangGraph, prompt engineering, agentic workflows, automated evaluation, calibration, bootstrap analysis, red teaming, SHAP
Data & Platforms: PostgreSQL/pgvector, Redis, Kubernetes, Docker, scikit-learn, pandas, NumPy, SciPy, Airflow, Databricks, Snowflake, BigQuery